

- Q.26 Explain the use of sealants.
- Q.27 What is the difference between Iron and Steel?
- Q.28 What are the different types of thermo plastics?
- Q.29 Explain creep and fatigue in a material.
- Q.30 Describe the critical points in iron carbon diagram.
- Q.31 What are the applications of Titanium sponge (Ti6Al4V) alloy?
- Q.32 What is use wrought Aluminum?
- Q.33 What is carburizing?
- Q.34 Name the light and heavy alloys used in aircrafts.
- Q.35 What are the precautions in storage of glues?

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Describe the properties and use of Ferrous materials in aircraft components?
- Q.37 How is heat treatment of carbon steels done? Mention the salient feature of Titanium and Inconel alloys.
- Q.38 Describe the various non-metallic materials used in aircraft constructions?

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4th Sem / Aircraft Maintenance Subject:- Aircraft Materials and Material Science-I

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What is not a type of wood?
- a) Mahogany b) Birch
c) Wrought d) Cedar
- Q.2 The typical density of Balsa wood is _____ Kg/m³
- a) 1000 b) 10
c) 160 d) 10000
- Q.3 Casehardening refers to _____ the material, after which necessary heat treatment is done to it.
- a) annealing b) critical range
c) carburizing d) hardening
- Q.4 Tires of aircraft are manufactured using?
- a) PVC b) CPVC
c) Rubber d) Charcol

- Q.5 Door seals of an aircraft are made of?
- a) Natural rubber b) Neoprene
 - c) Raw rubber d) Silicon rubber
- Q.6 Which of the following is not a common property for all aluminium alloys?
- a) Low strength b) Heat treatable
 - c) Easy for machining
 - d) Readily joined by welding
- Q.7 Which of the following is not a heat treatment?
- a) Brazing b) Annealing
 - c) Tempering d) Carburizing
- Q.8 Which of the following has highest carbon %age?
- a) Wrought Iron b) Mild steel
 - c) High carbon steel d) Cast Iron
- Q.9 What are the wheels of a landing gear made of?
- a) Solid rubber b) Silicon rubber
 - c) Titanium alloy d) Thick rubber
- Q.10 Is the standardization of materials useful/required?
- a) No, it is harmful to the aircraft
 - b) Required but not useful
 - c) Useful and required
 - d) Not Required but useful

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What are grain defects in woods?
- Q.12 What is the fabric type used in aircrafts?
- Q.13 What are the different types of steels?
- Q.14 What are the thermosetting plastics?
- Q.15 Why heat treatment is done?
- Q.16 What is the use of neoprene?
- Q.17 Where are reinforcing tapes used?
- Q.18 What is age hardening?
- Q.19 What is the main additive in Inconel?
- Q.20 What is Pickling?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Describe the various grain defects in woods.
- Q.22 What are the various plastics and their properties in brief?
- Q.23 Why and where synthetic rubber is used?
- Q.24 State the methods used for heat treatment of steel.
- Q.25 What are the requirements of dopes?